

Data assessment and preparation

Initial assessment

Player-level dataset for NBA Draft classes 2000-2026

2,347 players, 51 attributes

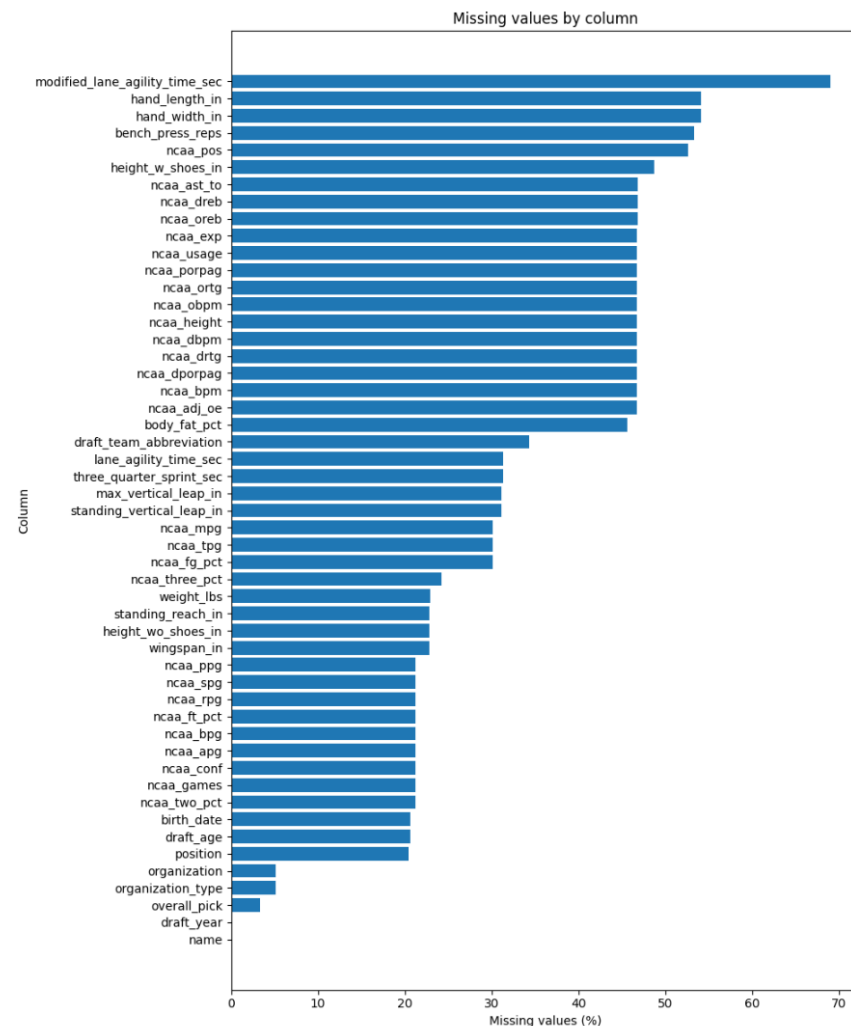
2026 players included without final draft labels

Main data issues

- Missing values are systematic, not random
- Combine data missing for some prospects
- NCAA statistics absent for international / non-college players
- Some advanced NCAA metrics had scale inconsistencies

Final preprocessing decisions

- Target transformed to 5 draft outcome groups
- NCAA-only scope: 1,773 historical / 74 in 2026 pool
- Leakage, sparse and inconsistent features removed
- Position-wise median imputation, one-hot encoding, scaling



Reasoning

Exact pick prediction is noisy and context-dependent. Broader outcome groups are more stable and easier to interpret.

Modeling and evaluation

Modeling setup

Train/validation experiment on historical NCAA-based data

Models tested:
Dummy Classifier, Logistic Regression, Gaussian Naive Bayes,
Linear SVM, Decision Tree, Random Forest

Main metric: macro F1-score (unbalanced classes)

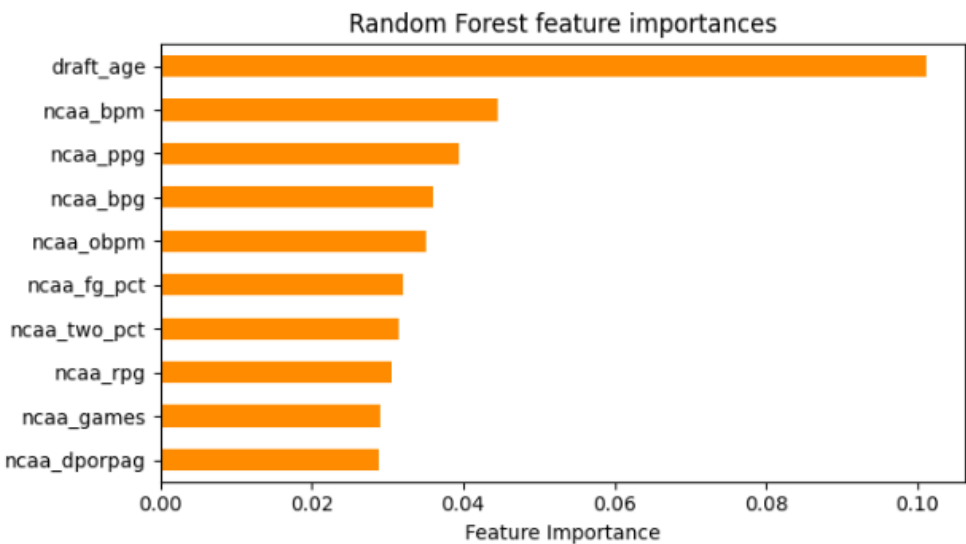
model	macro_f1	balanced_accuracy
Dummy Classifier	0.102	0.200
Logistic Regression	0.437	0.483
Gaussian Naive Bayes	0.080	0.202
Linear SVM	0.390	0.448
Decision Tree	0.424	0.425
Random Forest	0.530	0.522

Selected final model for final 2025 test

Random Forest with

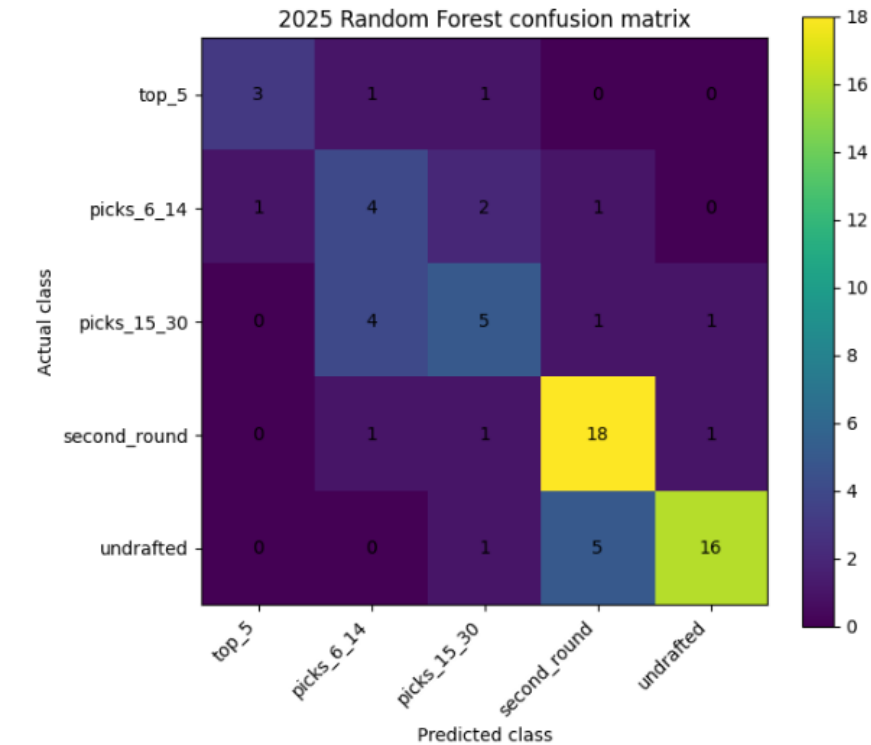
- n_estimators=300,
- max_depth=None,
- min_samples_leaf=5,
- max_features=sqrt,
- class_weight=balanced_subsample

Selected after hyperparameter tuning and cross-validation on 2000-2024 data

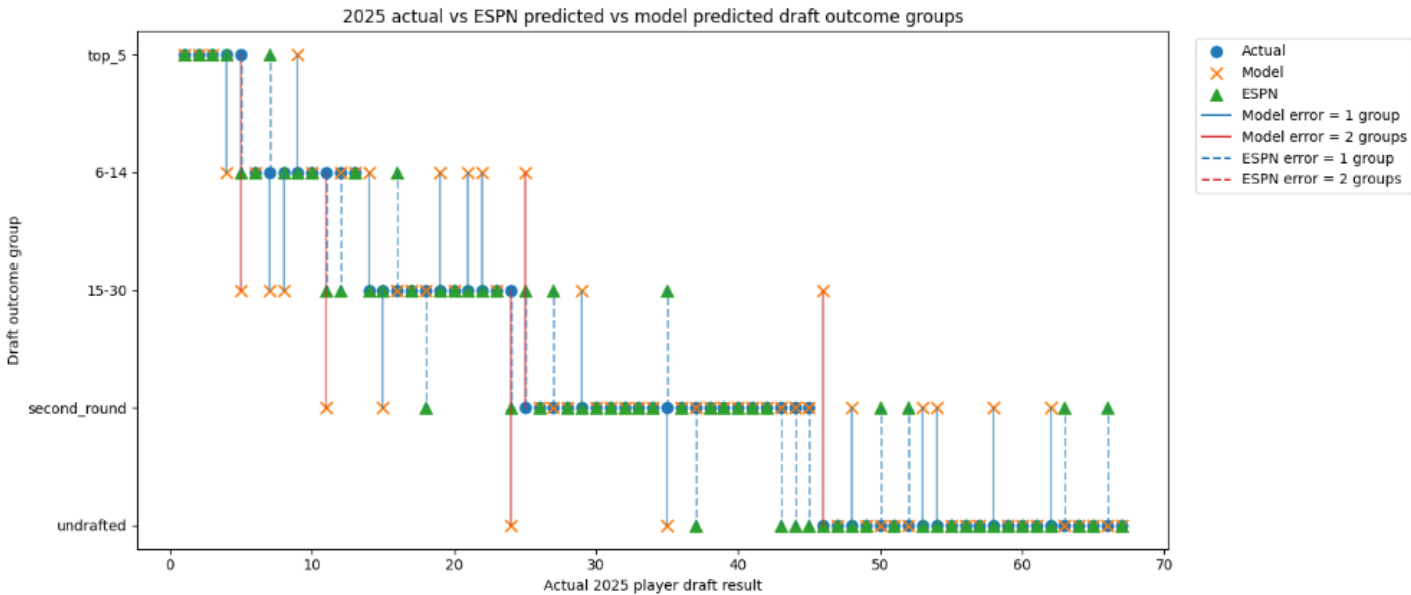


2025 test prediction

Final 2025 test: train on 2000-2024,
test on 2025 draft class



Comparison with expert predictions - ESPN mock draft from June 25, 2025



Main finding

The model was reasonably close to the expert baseline, but ESPN performed better overall. The largest differences appeared in context-heavy prospects, while the model still identified several less-publicized second-round players.

Method	Accuracy	Macro F1	Balanced acc.
Random Forest	0.687	0.634	0.628
ESPN mock	0.731	0.727	0.727
Model vs ESPN	0.507	0.492	0.491

2026 prospective prediction and expert comparison

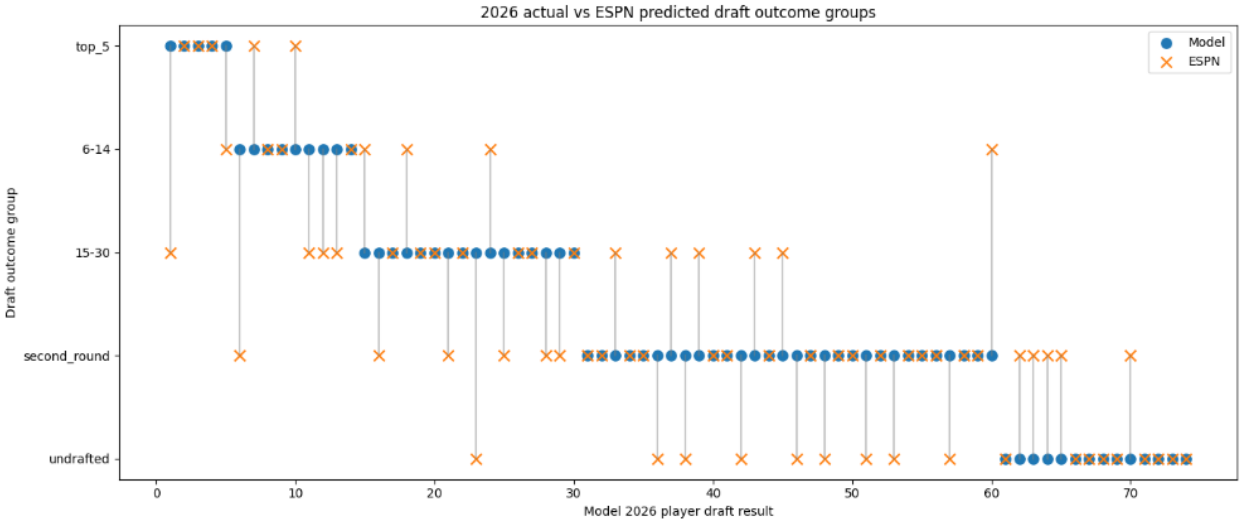
2026 prediction

Final model trained on all known historical outcomes from 2000-2025
2026 class is a prediction pool, not a test set
Raw model output first produced too many top_5-like profiles

Post-processing

To make the output more realistic, raw probabilities were converted into a draft-board ranking. Fixed draft capacities were then applied: 5 top_5, 9 picks_6_14, 16 picks_15_30, 30 second_round.

Model vs ESPN agreement	Value
Macro F1	0.503
Balanced accuracy	0.509



Largest disagreements

Allen Graves: model top_5, ESPN picks_15_30
Amari Allen: model picks_6_14, ESPN second_round
Jacob Cofie: model picks_15_30, ESPN undrafted
Yaxel Lendeborg: model second_round, ESPN picks_6_14

Discussion, deployment and conclusion

Conclusions

NBA Draft outcomes can be predicted to a meaningful degree from structured pre-draft data when the target is formulated as broad outcome groups. The Random Forest model provides an interpretable statistical baseline and a prospective 2026 prediction.

Limitations

No scouting reports, team needs, workouts, interviews, medical information or draft rumors

NCAA-only scope excludes several international prospects

Classification is not the same as a true constrained ranking task

Deployment idea

A GitHub Pages dashboard shows the 2026 model prediction and ESPN comparison. After the draft, a script will add actual results and update confusion matrices and visualizations without retraining the model.



Final result

The model does not predict draft picks any better than actual experts, but it comes close to their performance despite a huge number of limitations.